



LAB REPORT: Structures in C++ (Lab 13)



Title: Structure Declaration, Initialization and Access in C++



1. Structure Declaration and Initialization (Person)



Objective

To declare a structure named `Person`, initialize its members, and display values using dot notation.



Theory

A **structure (struct)** in C++ is a user-defined data type that allows grouping different data types under one name.

- Dot operator (.) is used to access structure members.
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Program Code

```
#include <iostream>
using namespace std;

struct Person {
    string name;
    int age;
    string address;
};

int main() {

    // Declare and initialize structure variable
    Person person1 = {"Ali Ahmed", 21, "Peshawar"};

    // Display values
    cout << "Person Details:" << endl;
    cout << "Name: " << person1.name << endl;
    cout << "Age: " << person1.age << endl;
    cout << "Address: " << person1.address << endl;

    return 0;
}
```



Explanation

- `struct Person` → structure created
 - `person1` → structure variable
 - `{}` → values initialized
 - `person1.name` → accessing member using dot operator
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Expected Output

Person Details:
Name: Ali Ahmed
Age: 21
Address: Peshawar

Conclusion

This program shows how to declare and initialize a structure and access its members using dot notation.

2. Structure Declaration and Initialization (Student)

Objective

To create a `Student` structure, initialize it, and display its values.

Program Code

```
#include <iostream>
using namespace std;

struct Student {
    string name;
    int age;
    char grade;
};

int main() {

    // Initialize structure variable
    Student student1 = {"Sara Khan", 20, 'A'};

    // Display values
    cout << "Student Details:" << endl;
    cout << "Name: " << student1.name << endl;
    cout << "Age: " << student1.age << endl;
    cout << "Grade: " << student1.grade << endl;

    return 0;
}
```

Explanation

- `Student` structure contains 3 members
 - `student1` stores data
 - dot operator accesses each field
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Expected Output

```
Student Details:
Name: Sara Khan
Age: 20
Grade: A
```

Conclusion

This program demonstrates structure initialization and accessing members in C++.

3. Accessing Structure Elements (Book)

Objective

To declare multiple structure variables and access their members.

Theory

Structures can store multiple variables of the same type and different types.
Multiple objects can be created from one structure.

Program Code

```
#include <iostream>
using namespace std;

struct Book {
    string title;
    string author;
    float price;
    int pages;
};

int main() {

    // Declare and initialize two structure variables
    Book book1 = {"C++ Basics", "John Smith", 450.5, 300};
    Book book2 = {"Data Structures", "Mark Lee", 600.0, 500};

    // Display book1 details
    cout << "Book 1 Details:" << endl;
    cout << "Title: " << book1.title << endl;
    cout << "Author: " << book1.author << endl;
    cout << "Price: " << book1.price << endl;
    cout << "Pages: " << book1.pages << endl;

    cout << endl;

    // Display book2 details
    cout << "Book 2 Details:" << endl;
    cout << "Title: " << book2.title << endl;
    cout << "Author: " << book2.author << endl;
    cout << "Price: " << book2.price << endl;
```

```
    cout << "Pages: " << book2.pages << endl;

    return 0;
}
```

Explanation

- Book structure defines book properties
- book1 and book2 are two objects
- each object stores different data
- dot operator used to access members

Expected Output

Book 1 Details:

Title: C++ Basics

Author: John Smith

Price: 450.5

Pages: 300

Book 2 Details:

Title: Data Structures

Author: Mark Lee

Price: 600

Pages: 500

Conclusion

This program shows how multiple structure variables can store and display different data using a single structure type.

Final Summary

Task	Concept Used	Key Idea
Person Structure	Initialization	Single object
Student Structure	Member access	Dot operator
Book Structure	Multiple objects	Reusability